AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS X EXAMINATION

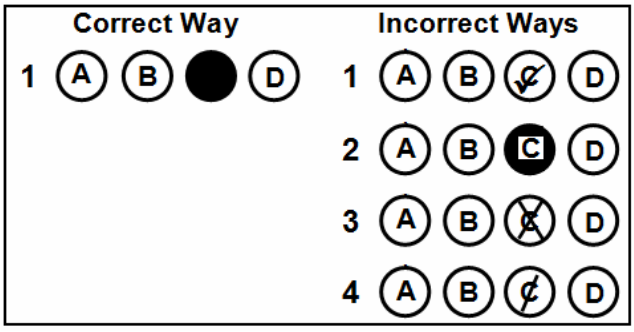
MAY 2015

Computer Science Paper I

Time: 35 minutes Marks: 25

INSTRUCTIONS

1. Read each question carefully.
2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.
3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 25 only.
4. In each question there are four choices A, B, C, D. Choose ONE. On the answer grid black out the circle for your choice with a pencil as shown below.



Candidate's Signature

5. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new circle.
6. DO NOT write anything in the answer grid. The computer only records what is in the boxes.

1. A positive impact of computer technology is that

   A. people are addicted to social networking websites.
   B. there is a growing trend of paper-less environment.
   C. our lives are too dependent on technology.
   D. computers are replacing people at work.

2. URL is a unique address for every website on the internet. URL stands for

   A. Uniform Record Locator.
   B. Uniform Resource Locator.
   C. Uninterrupted Record Locator.
   D. Uninterrupted Resource Locator.

3. Which of the following is/ are required for accessing internet?

   I. Modem
   II. Email id
   III. Web browser

   A. I only
   B. I and II
   C. I and III
   D. II and III

4. Which of the following is/ are the destructive effect(s) of a computer virus?

   I. Hard disk failure
   II. LCD failure
   III. Operating system failure

   A. I only
   B. I and II
   C. I and III
   D. II and III

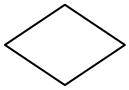
5. Which of the following can be implemented in both hardware and software, or a combination of both, to prevent unauthorized access?

   A. User login
   B. Antivirus
   C. Firewall
   D. Trojan

6. Which of the following is CORRECT about an algorithm?

   A. It consists of different graphical symbols.
   B. It is the simplified text version of a program.
   C. It consists of computer executable statements.
   D. It is too dependent on specific programming language.

7. Which of the following flow chart symbols is used for off-page connector?

   A.
   B.
   C.
   D. 

8. $z = a * b - c * d$

   Typing the above formula for calculating product of four numbers is an example of

   A. logical error.
   B. syntax error.
   C. execution error.
   D. data error.

9. Which of the following provides the user with the guidelines for installation and informs about the features of a software product?

A. Debugger
B. Documentation
C. Flow chart
D. Pseudo code

10. Which of the following is NOT an executable command in GW BASIC?

A. SYSTEM
B. KILL
C. RUN
D. REM

11. Which of the following is a numeric constant in GW BASIC?

A. $10,000\,(2xy)/(2x)$
B. $10000.0\,(2*x*y)\setminus(2*x)$
C. $\text{Rs.}10,000\,(2*x*y)/(2*x)$
D. $\$10000\,(2*xy)/(2*x)$

12. In GW BASIC, the expression $C = 2\pi R$ is written as

A. C = 2[22/7]R
B. C = 2*[22/7]*R
C. C = 2(22/7)R
D. C = 2*(22/7)*R

13. The statement used for multiple branching is

A. FOR… NEXT
B. ON…. GOTO
C. PRINT USING
D. READ – DATA

14. Which of the following is used to execute a statement according to a given condition?

A. WHILE WEND
B. IF ELSE
C. GOSUB
D. READ DATA

15. Consider the following GW BASIC statement.

10 FOR A = 5 TO 0 STEP X

Which of the following values is valid for the variable X?

A. – 5
B. 0
C. 1
D. 5

16. Consider the following GW BASIC program.

```
10 CLS
20 FOR A = 3 TO 4
30 PRINT A^2,
40 NEXT A
50 END
```

The output of the given program will be

A. 3 4
B. 6 8
C. 9 16
D. 12 32

17. Which of the following generates nested loops in GW BASIC?

A. WHILE WEND within FOR NEXT
B. ON ERROR GOTO within FOR NEXT
C. GOSUB within FOR NEXT
D. READ DATA within FOR NEXT

18. Which of the following is a valid subscript of array element in GW BASIC?

A. S(3)
B. S[3]
C. S{3}
D. S|3|

19. The output of the GW BASIC statement PRINT FIX(-99.65) will be

A. – 100
B. – 99
C. 99
D. 100

20. In GW BASIC, the statement SCREEN 0 is used for

A. high resolution graphics mode.
B. medium resolution graphics mode.
C. text mode only.
D. numeric mode only.

21. The output of the GW BASIC statement PRINT 2 ^ 2 + 2 * 2 will be

A. 8
B. 12
C. 16
D. 32

22. The output of the GW BASIC statement PRINT (1 OR 1) + (1 AND 1) will be

A. 22
B. 11
C. 2
D. 1

23. Consider the following GW BASIC program.

```
10 CLS
20 GOSUB 100
30 END
100 PRINT (23 + 24 + 45.0 + 90)/4
110?
```

Which of the following statements should be inserted in line 110?

A. KILL
B. RESET
C. RETURN
D. STOP

24. If A = 0, then $\overline{A}.A$ is equal to

A. – 1
B. 0
C. 1
D. 10

25. The simplified form of Boolean expression $\overline{X}.\overline{Y} + X.\overline{Y}$ will be

A. $X.\overline{Y}$
B. Y
C. X
D. $\overline{Y}$

END OF PAPER